Deterministic Relative Certificates for Grammar-Conditioned Distribution Drift

Gabriel Jota Lizardo Gabriel Barbosa da Fonseca

Research draft, 12 October 2026

Abstract

Structured diffusion decoders often condition a frozen product prediction on a recursive grammar. Comparing two such predictions requires measuring their conditioned distributions, rather than comparing independent token marginals. We give an integer implementation of planar support compression with $O(\eta^{-1/2})$ generators and near-linear construction cost. Combined with known admissible-gate propagation, this yields deterministic rational relative total-variation certificates on unambiguous event forests, with polynomial bit complexity and no positive-mass or positive-distance floor. For ordinary products the arithmetic bound is $O(nq\sqrt{n/\varepsilon}\log(q+2))$, improving the precision exponent of the stated deterministic bounds examined here. A classical chart construction gives a full-input polynomial corollary for trimmed LL(1) grammars and original token identities, with precision depending on emitted length and same-span chain height. For full-state Markov laws the bound is $O(nq^2\sqrt{n/\varepsilon}\log(q+2))$. We separate these quantitative claims from the known geometric, parsing and admissible-gate principles. A positive rational JSON family shows that all named inexpensive upper bounds can have an unbounded relative gap; an exact specialized counter still wins on that family. A complete independent 360-row diffusion-head audit fails its predeclared precision gate. A common final-query optimization yields only one 2x developmental pair in a second complete replay. These negative and limited observations are retained: the contribution is a proposed mathematical audit bound, without a faster-generation, semantic, EPIC or publication-acceptance claim.

1 Problem and quantitative contribution

Let $S_1, \dots, S_n$ be finite domains of original token identities. Each token has a fixed byte emission. A canvas and an unambiguous grammar specify a common valid set $\mathcal{F} \subseteq \prod_i S_i$. For two nonnegative rational tables p_i, q_i with row sums at most one, let

$$Z_p = \sum_{y \in \mathcal{F}} \prod_i p_i(y_i), \qquad P(y) = \frac{\prod_i p_i(y_i)}{Z_p},$$

and define Z_q, Q analogously. We require $Z_p, Z_q > 0$; otherwise this comparison is undefined. Zeros, token aliases, fixed positions and rare valid events are allowed. The target is a rational interval

$$L \le \mathrm{TV}(P, Q) \le U \le (1+\varepsilon)L, \qquad 0 < \varepsilon \le 1. \tag{1}$$

This is the frozen, grammar-conditioned product law of a step, not the learned joint law of a complete neural trajectory. No EOS is admitted here.

Masked diffusion models motivate parallel structured sampling [9]. Grammar conditioning and semiring parsing are antecedents [11, 10]; this paper audits the distributional effect of numerical tables used in that operation. EPIC [12] instead accelerates grammar-constrained commitment. These tasks are distinct; computing a tighter drift certificate does not establish a faster or more accurate EPIC replacement.

TV as a support query, admissible-gate composition and deterministic relative approximation are established principles [1]. Classical convex approximation gives the planar exponent [3, 4]; constructive bit-model zonotope approximation is also an antecedent, not a first capability. Our construction retains original integer polygon endpoints; its generators have no new denominators, and its nearest-edge queries use exact integer comparisons. These properties provide the explicit arithmetic and bit bounds below. We do not claim a new geometric approximation rate or a TV lower bound.

Feng, Liu and Liu's product FPTAS has stated arithmetic cost [2, Theorem 1]

$$O(qn^2\varepsilon^{-1}\log q\log(n/(\varepsilon\,\mathrm{TV}))).$$

Its $(1-\varepsilon)$ convention converts to (1) by using error $\varepsilon/(1+\varepsilon)$. Fu's paired product corollary gives $\widetilde{O}(nq(n/\varepsilon)^4)$ and already removes numerical TV dependence in an extended real-arithmetic model [1, Corollary 1.4]. Our sharper product bound uses precision exponent $1/2$ and polynomial-bit integers. Comparisons are of published upper bounds, not lower bounds on competent alternative implementations or guarantees of physical speed. Bhattacharyya, Feng and Srivastava [8, Sections 2–3] restate the Feng bound and simplify its extension-distance proof; they already credit its polynomial bit complexity. Integer computability alone is therefore not our claimed improvement. Fu already removes the positive-TV dependence.

Randomized inference-based relative approximation is also an antecedent [5]; a derivation encoding supplies a relevant alternative, as discussed in Section 6. Randomization may be preferable when a probabilistic failure bound is acceptable. The finite numerical audit does not compare those algorithms' runtimes; its independent precision gate failed.

2 An exact integer planar core

For a finite multiset G of integer pairs, put

$$F_G(w)=\sum_{g\in G}|g\cdot w|,\qquad Z(G)=\sum_{g\in G}[-g,g].$$

Thus F_G is the support function of the centrally symmetric polygon $Z(G)$.

Theorem 1 (Integer endpoint compression)**.** *Given m integer pairs and integer $k\ge 1$, one can construct a multiset C of at most $64k$ integer pairs such that, for every $w\in\mathbb{R}^2$,*

$$(1-(16k^2)^{-1})F_G(w)\le F_C(w)\le F_G(w). \tag{2}$$

The construction uses $O(m\log(m+2))$ exact arithmetic operations, or $O(m)$ if the nonzero, noncollinear generators are already in canonical angular order. The returned core preserves that order. Each output coordinate is a signed sum of input coordinates. Rank zero or one is compressed exactly. Intermediate integers have polynomial input bits.

Proof. Orient nonzero generators into a half-plane and sort by determinant. Starting at minus their sum and walking by twice each generator, then its opposite, enumerates $Z(G)$. Every polygon vertex has the same coordinate parities. Redundant collinear vertices may be removed. For rank two choose a vertex u of maximum squared Euclidean norm and a vertex v of maximum $|\det(u,v)|$, reversing v to make $D=\det(u,v)>0$. If $T=[u\ v]^{-1}$ and $K=TZ(G)$, then

$$\mathrm{conv}(\pm e_1,\pm e_2)\subseteq K\subseteq[-2,2]\times[-1,1]. \tag{3}$$

Indeed the coefficient $\beta=\det(u,x)/D$ is at most one in absolute value. Writing $x=\alpha u+\beta v$, maximality of $\|u\|$ gives $|\alpha|\|u\|\le\|x\|+\|v\|\le 2\|u\|$. The inner inclusion follows from the four chosen vertices. Hence $h_K(w)\ge\|w\|_\infty$.

Take the $32k$ points of spacing $1/k$ on the boundary of $[-4,4]^2$. For each, find a closest polygon edge and retain both of its endpoints and their opposites. Let R be their convex hull. Clearly $R \subseteq K$. For a unit supporting normal w, let x maximize $w \cdot x$ on K. The ray $x + tw$ reaches the outer square at p, with $t \geq 2$. A grid point q on that side has $\|q - p\| \leq 1/(2k)$. Let y be the projection of q onto K. Projection optimality gives

$$(q-y)\cdot(x-y) \leq 0, \qquad t\,w\cdot(x-y) \leq -\|x-y\|^2 - (q-p)\cdot(x-y) \leq \frac{\|q-p\|^2}{4}.$$

Because y belongs to a retained edge, it is in R. Thus $h_K(w) - h_R(w) \leq 1/(32k^2)$. Since $h_K(w) \geq 1/2$ for unit w, homogeneity gives (2) for these two polygons.

Map the retained vertices back by T^{-1}. Their centered hull has $2r \leq 128k$ vertices $z_0, \ldots, z_{2r-1}$ in boundary order, with $z_{i+r} = -z_i$. Its generators are $b_i = (z_{i+1} - z_i)/2$, $0 \leq i < r$. They are integers by the common parity, and each is the sum of a consecutive run of oriented original generators. Their starting vertex is $-\sum_i b_i = z_0$, so they represent the hull exactly. One may return the original list if it is shorter.

If $m \leq k$ return the original ordered generators exactly, avoiding a large grid. Otherwise $k < m$. Polygon construction after sorting costs $O(m)$; constant-factor normalization uses two linear scans. Lemma 1 handles all outside queries together in $O(m + k) = O(m)$ time. Mark retained vertices and opposites in a Boolean array, then scan in original boundary order instead of sorting a new hull. Consecutive first-half edge differences are already cyclically ordered; canonical orientation introduces at most one wrap. Locate the minimum angle by positive determinant comparison and rotate the list. This costs $O(m)$ and yields ordered output. The only $O(m \log(m+2))$ work is the initial sort when the input is unordered. After scaling by kD, all geometry uses integer cofactors, determinants, squared segment distances and cross multiplication. No projection coordinate is output and no new rational denominator is introduced. □

Lemma 1 (Monotone outside sweeps)**.** *For a convex polygon strictly to the left of the outside query line, all queries on that line in increasing vertical order have exact nearest edges in linear total arithmetic time in the polygon and query counts. Quarter-turns handle the other three sides.*

Proof. The projection lies on the right boundary chain, from the rightmost lowest vertex to the rightmost highest vertex. Along this chain the second coordinate increases strictly after collinear removal. Write the boundary as $x = f(t)$, where f is concave and $4 - f(t) > 0$. Then

$$D_Y(t) = (4 - f(t))^2 + (Y - t)^2$$

is strictly convex. If $Y_1 < Y_2$ and t_1, t_2 are its respective minimizers, adding their two minimizing inequalities gives $(Y_2 - Y_1)(t_2 - t_1) \geq 0$. Hence the projection's vertical coordinate never decreases. Distances minimized over consecutive segment intervals decrease then increase, with a possible tie at the shared minimizing vertex. Start a pointer at the first segment. For each successive query advance while the next segment distance is no larger; choose the last edge in a tie. The pointer cannot move backwards and advances at most the chain length. Each query and each advance costs constant exact squared-distance comparisons by integer cross multiplication. Four rotated chains and their preparation cost $O(m + k)$ arithmetic operations and linear space. □

The $O(\eta^{-1/2})$ order is classical worst-case optimal for approximating *all* planar support directions: an inscribed $2r$-vertex polygon has an angular gap at least π/r, and a circle loses order r^{-2} support at that gap's midpoint. Fine rational polygons approximate the circle. This is not a lower bound for computing one TV query.

3 Event forests, relative error and bit cost

Definition 1. *An admitted event forest is an acyclic sum/product circuit with a fixed set of original positions, or scope, at each node. A leaf lists distinct choices at one position, or is an*

empty-scope unit/zero. Product scopes are disjoint and their events form a Cartesian product. Sum alternatives have equal scopes and disjoint assignment events. The root represents each member of $\mathcal{F}$ once. Productions have neutral weights. Empty-scope nodes reduce to unit/zero. There is no common-v-tree requirement.

For each head independently let D_i clear all denominators at position i, and let $a_i(v) = D_i p_i(v)$; use separate denominators and b_i for q. A position denominator may be the product of its atom denominators: $D_i = \prod_v \operatorname{den}(p_i(v))$. Then $a_i(v)$ uses one exact integer quotient/multiplication per atom, and $\log D_i$ is bounded by the position's input bits. This clears denominators in linear arithmetic work, without counting a bit-length-dependent GCD as one primitive. A largest-denominator divisibility shortcut applies to dyadic rows. The measured reference retains LCMs/Fraction and charges their actual time; this alternative is for the complete arithmetic theorem. Return an unreduced integer fraction pair when counting constant output arithmetic; reducing it also has polynomial bit cost. Compute the exact positive integer root masses A, B by ordinary inside addition/multiplication. They are not inferred from compressed generators. At a leaf keep pairs $(a_i(v), b_i(v))$; at a sum concatenate child lists; at a product take all componentwise products of child pairs. Compress each internal gate by Theorem 1. Core pairs may be signed: they represent norms, never probabilities or a token sampler. The rounded head Q remains the actual alternative law. Exact unary unions and singleton diagonal maps may pass through without compression. If omitted from the error count, they contribute zero error; all experimental methods receive these same shortcuts.

Let d be the maximum number of compression gates in an expanded root derivation: a sum takes the maximum child count plus one; a binary product adds both child counts plus one. Use $d \geq 1$. Set $k = \lceil \sqrt{d(1+\varepsilon)/(16\varepsilon)} \rceil$, computed by integer square root and comparisons. At the root return

$$L = \frac{\sum_{(x,y)\in C} |Bx - Ay|}{2AB}, \qquad U = \min\{1, (1+\varepsilon)L\}. \tag{4}$$

Theorem 2 (Deterministic rational forest certificate)**.** *The algorithm satisfies* (1)*, including TV zero exactly. Let H count explicit nodes and wires. Then $d \leq H$. With bounded fan-in and bounded exact leaf lists, its core arithmetic cost and storage are*

$$O\big(H(d/\varepsilon)\log(d/\varepsilon+2)\big), \qquad O\big(H\sqrt{d/\varepsilon} + d/\varepsilon\big).$$

For general leaf lists of maximum size V, set $M = 64k$, $K_0 = \max(V, M)$ and $J = HK_0 + K_0^2$. A conservative full bit bound is

$$\operatorname{poly}(\ell) + O\big(HJ\log(J+2)(\ell + \log(J+k+2))^2\big),$$

where ℓ is total input bit length. No conditioning-mass or TV floor is needed. Preparation of an input representation is charged separately.

Proof. If two child cores preserve all query norms from below by factors α and β, disjoint sum preserves their smaller factor. For a product, fix each right generator and query the left norm with its componentwise product with w. Sum the inequalities, then do the same for the right norm using the retained left generators. This gives lower factor $\alpha\beta$ and upper factor one, even with signed retained pairs. Induction gives $(1 - 1/(16k^2))^d F_{\text{exact}} \leq F_C \leq F_{\text{exact}}$. Bernoulli's inequality and $16k^2 \geq d(1+\varepsilon)/\varepsilon$ imply a lower factor at least $1 - \varepsilon/(1+\varepsilon) = 1/(1+\varepsilon)$. At query $w = (1/A, -1/B)$ the exact norm is $2\,\mathrm{TV}(P, Q)$, proving the interval.

A nonempty scoped node cannot repeat in a derivation: its lowest common ancestor would be a product whose two disjoint scopes contain that same nonempty scope. Acyclicity prevents repetition along a chain. This proves $d \leq H$; repeated-squaring arithmetic DAGs are excluded by the scope contract.

Distinct-event mass at scope I is at most $\prod_{i\in I} D_i$, since each input row sums to at most one. Evaluate core containment on either coordinate axis: the sum of absolute coordinates is at most that exact head mass. This invariant also bounds product/union intermediates before compression. Each coordinate therefore has $O(\ell)$ bits. Polygon vertices, cofactors, scaled grid points and squared-distance comparisons have $O(\ell + \log(J + k + 2))$ bits, up to constant factors. Standard integer multiplication gives the displayed conservative bit bound. Initial LCMs, normalizers and final rational operations contribute polynomial input work. At most J candidates occur at a gate; Theorem 1 then bounds every compression. Under bounded arities and leaf sizes products have $O(M^2)$ rows and sums $O(M)$, giving the sharper expressions. □

Corollary 1 (Products)**.** *For two rational product measures on n domains of size at most q, the arithmetic cost is*

$$O(nq\sqrt{n/\varepsilon}\log(q+2))$$

plus input/scaling/normalizer work, with polynomial-bit arithmetic.

Proof. Start with $(1, 1)$. At position i, merge the diagonal images $(a_i(v)x, b_i(v)y)$ for every original v and every previous core pair (x, y). Compress once. Here $d \le n$ and the intermediate list has qM pairs; it does not form a wasteful general M^2 Cartesian product. Each nonnegative diagonal map preserves canonical angular order, including singular maps which collapse to one direction after omitting zero images. An exact q-way heap merge costs $O(qM\log(q+2))$, aggregates collinear images and supplies an ordered list to the linear compressor. No original probability event is discarded: the list represents a support norm, not a sampling law. The generic forest product may use signed diagonal maps and still has the preceding, different complexity bound. Normalizer masses are products of integer row sums. Apply the preceding norm induction and sweep bound. For an empty product TV is zero. □

Corollary 2 (Full-state Markov trajectories)**.** *For two rational laws on entire n-step trajectories of q-state inhomogeneous Markov chains, the arithmetic bound is*

$$O(nq^2\sqrt{n/\varepsilon}\log(q+2)),$$

including input/scaling/forward-normalizer work, with polynomial-bit integers.

Proof. Clear denominators with one multiplier for the WHOLE transition matrix at each position, separately for each law. A different multiplier per previous state would change path weights and is invalid. Maintain an ordered paired prefix core for each endpoint. To reach endpoint k, merge the q diagonal images of preceding endpoint cores, scaled by the two transition weights (j, k), and compress once. Preceding states define disjoint trajectory events. There are q endpoints and q streams per endpoint. The final endpoint union adds at most one compression, so $d \le n$. Exact forward masses and the preceding norm/bit induction prove the interval and cost. This concerns full trajectories, not observed-word marginals of a nondeterministic hidden model. □

The stated deterministic Markov upper bounds examined here are

$$O(n^2q^2\varepsilon^{-1}\log q\log(n/(\varepsilon\,\mathrm{TV}))) \quad\text{and}\quad \widetilde{O}(nq^2(n/\varepsilon)^4),$$

respectively [2, Theorem 2][1, Corollary 1.5]. The recurrence and heap merge are classical; the precision exponent is the comparative claim, not a lower bound on every implementation. The same proof applies to original-labeled deterministic layered graphs, but materializing a predictive stack graph can be exponential and expanding original edges can be worse than a compact token-group forest.

4 A polynomial original-word grammar corollary

The forest-input theorem is not a proof of a software normalizer. Here is a separate mathematical full-input construction. Trim unreachable and unproductive symbols of an LL(1) byte grammar. Keep unit/epsilon rules, replace terminals in longer bodies by preterminals, and right-binarize with fresh fixed-chain auxiliaries. This is linear in grammar symbol occurrences g and preserves derivations bijectively, without nullable-subset expansion.

Construct an acyclic byte-prefix trie per original token position with a separate final-byte closure edge for each original ID. Prefix tokens close to the next boundary while longer tokens continue within the current trie. Aliases have distinct closure edges; their probabilities occur once on those edges. Empty/unsupported emissions are invalid and keep their mass in the input table. Complete lattice paths and original assignments are bijective. Let V_b, E_b be lattice vertices/edges; these are bounded by total declared token-byte occurrences before sharing.

Theorem 3 (LL(1) source-input certificate)**.** *For the trimmed LL(1) source, original token domains and positive valid masses above, a deterministic rational certificate* (1) *is computed with total cost polynomial in* $g, V_b, E_b, \ell, 1/\varepsilon$*. No common parse tree or fixed maximum nesting depth is assumed.*

Proof. For a binary rule $A \to BC$ add same-span dependency $A \to B$ if C is nullable and $A \to C$ if B is nullable; add unit dependencies as well. Every dependency is a nullable-prefix left corner. A cycle collapses to source left recursion, excluded by LL(1); auxiliary chains alone cannot cycle.

Process chart cells (A, u, v) by increasing topological difference $v - u$ and then in dependency order. At $u = v$ use the unique empty derivation or zero. At positive spans include matching byte edges, unit children and binary splits at every $u \le w \le v$. Interior children have shorter spans; endpoint children use a nullable unit and an already computed same-span child. Hence the circuit is acyclic and has $O(gV_b^2)$ cells and $O(gV_b^3 + gE_b)$ alternatives.

Standard chart induction gives every path parse exactly once. LL(1) unambiguity makes alternatives disjoint as original assignment events, not just tree tags. Trimming ensures relevant nonterminal ambiguity cannot be hidden outside the start language. Vertex layers determine consumed closure slots, so sum scopes agree and product scopes partition. An empty scope stays inside one trie, with a unique path and unambiguous parse, and reduces to unit/zero. Theorem 2 applies. All source/lattice work is charged. This is classical parsing used for admission, not a new parsing algorithm. □

Let $B = \sum_i \max_{v \in S_i} |\,\mathrm{emission}(v)|$ over supported nonempty emissions, and let h be the acyclic same-span dependency height. For a root derivation with $b \le B$ terminal byte edges, contract maximal same-span positive-cell chains. They have at most $h + 1$ cells; the contracted full binary tree has $2b - 1$ nodes. At most an OR and a selected AND compression occur per positive cell; implicit empty subtrees are exact. Consequently

$$d \le 2(h + 1)(2B - 1) = O(gB).$$

For an already admitted unambiguous epsilon/unit-free CNF, direct binary-tree counting gives the sharper $d \le 3B - 2$. No ambiguity-preserving CNF conversion is silently assumed for arbitrary source input. Keep chart unions flat: an f-alternative union sorts $O(fM)$ rows. Including wires in H, the full core arithmetic upper is

$$O(H[(d/\varepsilon)\log(d/\varepsilon + 2) + \sqrt{d/\varepsilon}\log(H + 2)]).$$

The second term pays wide-OR sorting; binary conversion could introduce extra compression error. Grammar/lattice construction and normalizers are additional charged work. This is a written algorithm, not verification of the maintained normalizer or the measured materialized-stack compiler.

5 A software use: auditing numerical probability tables

If a certified upper bound U is below an application's error budget τ, replacing P by Q changes the expectation of *every* verifier with values in $[0,1]$ by at most τ. This is the standard operational meaning of TV; it requires no chosen semantic labels or favorable benchmark. For numerical table storage, nominal FP64-to-FP32 storage halves and FP64-to-16-bit storage quarters before metadata and transfers. We certify the distributional effect, not model throughput or parameter quantization.

Multiplying each row by a positive constant does not change its conditioned law: every valid assignment consumes each declared position exactly once, so the common product of row constants cancels in its normalizer. A sampler can therefore use integer-scaled rounded atom weights without carrying row normalization constants through sampling. This is elementary and not our algorithmic novelty; the certificate audits the resulting changed law.

A common continuation kernel cannot increase TV. For a finite iterative decoder with uniform per-reachable-history certificates τ_t, a coupling gives final trajectory distance at most $\sum_t \tau_t$. This consequence requires the same sampling/commitment policy. A confidence threshold that changes under rounding needs a separate policy-preservation argument; local head closeness alone does not prove trajectory closeness. One observed history also does not certify every possible future history.

Corollary 3 (Budgeted table substitution)**.** *Consider a finite-horizon decoder which samples a grammar-conditioned head and applies a common update rule. Keep its original policy inputs and change only the table used by the sampler. Start with budget $\tau \in [0,1]$. At each processed history, admit a substitute only with a valid certificate U no larger than the remaining budget, then deduct U; otherwise keep the original table and deduct zero. The resulting trajectory law differs from the original by TV at most τ, hence every final $[0,1]$-valued verifier changes in expectation by at most τ.*

Proof. While histories match, maximally couple the two sampler laws. Their disagreement probability at that step is at most the admitted U, also after contraction under the common update. The probability of any first disagreement is bounded by the expected sum of these certificates over the matching prefix. Each substituted trajectory has nonnegative remaining budget, so its full sum, and thus that prefix sum, is at most τ. The coupling inequality proves the conclusion. □

This standard coupling application does not require precomputing all future heads: admission enforces the invariant whenever a history is reached. Budget affects table admission only, not the common policy. Resource refusal or unsupported input retains the original table; a zero reference valid mass uses the same declared failure behavior. It does not automatically apply to native EPIC or to changed confidence thresholds. No iterative neural run of this wrapper, model quantization or generation-speed gain is measured here.

5.1 Tight decisions and a gap of the named cheap bounds

Given an externally chosen budget τ, accept table substitution when $U \le \tau$, reject it when $L > \tau$, and otherwise remain unresolved. The relative guarantee ensures acceptance whenever $\mathrm{TV} \le \tau/(1+\varepsilon)$ and rejection whenever $\mathrm{TV} > (1+\varepsilon)\tau$. These are elementary consequences, not an additional algorithmic discovery. Competent lower controls include exact original-token marginal TV, support losses, Hellinger and moment inequalities; 105 independent tiny word cases validate our reference intervals. No budget was selected from observed outcomes, and this decision operation has not been measured on the external heads.

Lemma 2 (Non-relative gap of the declared cheap controls)**.** *There are positive rational product laws on $n+1$ original Boolean tokens which render valid JSON arrays, for which the best of*

ideal Hellinger, both chi-square, sharp likelihood-range and sequential categorical-sampler coupling uppers is $\Omega(\sqrt{n})$ times the actual TV. Full/common-support variants coincide. This does not concern every TV algorithm or scalar formula.

Proof. Let integer $m \geq 4$, $n = m^2$, $\delta = m^{-4}$ and $a = m^{-3}$. At the first position $P(1) = a$, $Q(1) = 2a$; at each other position $P(1) = 1/2$, $Q(1) = 1/2 + \delta$. Projection gives TV $\geq a$. On the balanced block, $\chi^2(Q_{\text{dense}} \| P_{\text{dense}}) = (1 + 4\delta^2)^n - 1 \leq 8/m^6$, using the binomial expansion and its geometric upper. Product triangle inequality and Cauchy–Schwarz therefore give TV $\leq a + \sqrt{2}/m^3 < 3a$.

Sequential coupling has upper $1 - (1-a)(1-\delta)^n \geq 1 - (1-\delta)^n \geq 1/(2m^2)$ by the two-term Bonferroni bound. Chi-square tensorization gives forward value at least $a/(1-a)$ and reverse at least $a/[2(1-2a)]$; both capped half-square-root uppers exceed $1/(2m^2)$. Hellinger tensorization bounds its upper below by the first-coordinate value

$$\frac{\sqrt{a}}{\sqrt{2(1-a)} + \sqrt{1-2a}} > \frac{\sqrt{a}}{3} \geq \frac{1}{2m^2}.$$

The likelihood extrema are $l = [(1-2a)/(1-a)](1-2\delta)^n$ and $h = 2(1+2\delta)^n$. Since $h \geq 2$ and $1 - l \geq 1 - (1-2\delta)^n \geq 1/m^2$, the sharp range upper $(1-l)(h-1)/(h-l) \geq (1-l)/2 \geq 1/(2m^2)$. Thus every named ideal upper divided by TV is at least $m/6$. Fixed JSON array punctuation and original 'true'/'false' token choices make every assignment valid, with valid mass one and the same TV. □

This is an abstract proof object, not a model benchmark. Exchangeability allows an EXACT $O(n)$ binomial counter on this family: it is a better specialized algorithm and must not be replaced by exponential enumeration as a performance opponent. Six rational checks use that counter. The lemma shows that the listed cheap controls lack a near-relative guarantee; Feng and Fu already provide general near-relative algorithms, against which our comparison remains quantitative. No typical neural-head benefit follows.

5.2 Complete independent audit and an explicitly developmental replay

We froze ALL five eligible original-byte documents in the pinned public JSONTestSuite, 4/8/16 masked positions, FP32/FP16/BF16 probability-table rounding, both tolerances $1/2, 1/10$, and all four engines before new forwards. There are 15 actual full-vocabulary MDLM heads (50,258 original IDs), plus one warmup. Four documents are unseen by this TV development, not necessarily by model training or earlier project work. Whole-row rational normalization keeps zeroed and unsupported original IDs. The model/tokenizer are pinned and no parameters are updated; these are frozen-head audits, not iterative generation.

Exact normalizers are checked independently against preserved stack sources. The measured bridge retains original IDs, aliases and closure choices. Explicit stack preparation can grow exponentially; it is charged and does not implement the polynomial chart theorem above. All 360 rows conclude: 256 certificates and 104 resource refusals, no missing rows or computation errors. Shared refusals count for every affected method and receive no victory credit. All cases, tolerances and failures remain publicly recorded.

Controls include exact likelihood-range extrema, certified global/common Hellinger, both common-support chi-square directions with relative atom rounding, and full/common-support sampler overlap. For that last classical control, compute exact inside masses A_v, B_v and alternative masses a_t, b_t. Its lower bound on sampler agreement obeys

$$c_v = \sum_t \min(a_t/A_v, b_t/B_v) \prod_{u \in t} c_u, \qquad \text{TV} \leq 1 - c_{\text{root}}.$$

Downward dyadic evaluation certifies this upper; support corrections are charged. This stronger baseline eliminates most of the earlier developmental gap. The unchanged independent predicate

requires a candidate upper at least 2x tighter than the best cheap upper, at $\varepsilon = 0.1$ with ordered cores. It FAILS in all 36 unseen pairs. Ideal-control-rounding robustness adds no wins. The best near-threshold ratio, 1.98850, is not rounded into success. The entire audit takes 3,750.830 seconds wall/CPU, including input, preparation, both normalizers, controls, core work and process management; model capture/load/warmup are recorded separately.

A classical final-query shortcut, supplied to ALL four engines, streams $|Bx - Ay|$ directly at the root instead of materializing/compressing its final union. Exact propagation of each compressor's rational lower factor gives $U = \min(1, L/\beta)$ with $\beta \geq 1/(1+\varepsilon)$; factor arithmetic and root work are charged. All 300 tiny original-word/engine cases pass before replaying ALL old 15 heads/360 configurations under the same guards. This second campaign also has 256 certificates and 104 refusals. 1 of 45 DEVELOPMENT pairs exceeds the old 2x diagnostic (2.02283x), also against ideal control-rounding enclosures. It is not independent confirmation or a successful latency gate; the failed original gate is preserved. Table 1 reports every ordered 0.1 pair from both campaigns.

Document / masks		FP32		FP16		BF16	
		V25	V26	V25	V26	V25	V26
string_uEscape	4	0.95384	0.99233	1.05476	1.09733	1.00695	1.04758
string_uEscape	8	1.13596	1.15096	1.42437	1.44318	1.12943	1.14434
string_uEscape	16	ref.	ref.	ref.	ref.	ref.	ref.
object_extreme_numbers	4	0.91536	0.96566	0.93152	0.98271	0.92940	0.98048
object_extreme_numbers	8	1.17544	1.20032	1.44486	1.47544	1.21684	1.24259
object_extreme_numbers	16	ref.	ref.	ref.	ref.	ref.	ref.
string_accepted_surrogate_pairs	4	1.12696	1.17244	1.06165	1.10449	1.12526	1.17066
string_accepted_surrogate_pairs	8	1.13506	1.15005	1.16217	1.17752	1.13521	1.15020
string_accepted_surrogate_pairs	16	ref.	ref.	ref.	ref.	ref.	ref.
object_string_unicode	4	1.06731	1.10267	1.00336	1.03660	1.10535	1.14197
object_string_unicode	8	1.12948	1.14889	1.57133	1.59833	1.13212	1.15157
object_string_unicode	16	ref.	ref.	1.98850	2.02283	ref.	ref.
object_long_strings	4	0.90984	0.95984	0.91013	0.96014	0.90950	0.95948
object_long_strings	8	1.05108	1.06080	0.99036	0.99952	0.95436	0.96318
object_long_strings	16	ref.	ref.	ref.	ref.	ref.	ref.

Table 1: All 45 ordered $\varepsilon = 0.1$ pairs per campaign: best cheap upper divided by candidate upper. 'ref.' is an explicit resource refusal. V25 is independent only for the four documents after the first; V26 reuses ALL old heads and is DEVELOPMENT. Values rounded for display; predicates use exact rational values. All360 records per campaign, including other methods and tolerances, remain in the immutable artifact. No semantic/speed gate.

Inputs are reproducible from stored rational probability tables; raw new logits/probability arrays are optional local inputs, not published in Git. Their hashes and exact-softmax/original-byte replay are recorded, but hashes alone are not complete offline reproduction of neural origin. Small tests and all-query oracles validate implementations; they do not establish novelty, human proof review, editorial acceptance or native EPIC superiority.

6 Related alternatives and exact claim boundaries

Fu's general admissible-gate theorem already permits the disjoint union and componentwise product operations of our forest [1]. Therefore the generic existence of deterministic approximation on an admitted forest is a specialization, rather than our main discovery. The smaller integer core and resulting quantitative bound are the claim of this draft. Specific priority is based on the primary results inspected as of this draft's date, not an exhaustive-literature guarantee.

Domain reduction for smooth decomposable probabilistic circuits requires a common v-tree in its current theorem; determinism is not required [6]. A parse forest can have differing product

splits. The ACTUAL explicit-stack prototype is a fixed-slot chain compatible with a common tree, so its examples do not establish a capability unavailable to that randomized theorem. This is not an encoding-independent separation, and the more general admissible-gate framework remains a competitor. Ordinary exact enumeration is preferable for small valid sets. Certified sampler-overlap, Hellinger/KL or range bounds are preferable when their cheaper upper guarantees suffice.

The ICML inference reduction [5] also deserves a competent grammar encoding. Normalize each reachable alternative by exact inside masses; activate the children selected by their parents, and give unused nodes a canonical inactive value. Decomposability precludes repeated nonempty nodes in a derivation. Complete choice vectors and valid assignments are bijective. Boolean activation gadgets avoid exponential CPT descriptions for shared nodes. For the reduction's diagonal and topological-prefix queries, minimum local conditional probabilities and clamped already active choices are ordinary weighted inside operations. A visited node's activation is determined by the prefix, so arbitrary future activation evidence is unnecessary. This is a classical randomized alternative, not a newly claimed reduction or a measured implementation. Its failure parameter and ε^{-2} sample dependence differ from the deterministic bound here. Recent linear-time randomized product algorithms are also stronger choices in their setting [7].

EPIC and commitment selectors perform different operations. This manuscript neither evaluates EPIC's complete law nor claims generation superiority over it. Exact token lifting, geometric queries and selected Lean specifications elsewhere in the project do not establish academic novelty by themselves. No formal proof of this complete geometric theorem or Python implementation, independent human review, training improvement or editorial acceptance is claimed.

7 Reproducibility and unresolved obligations

All source/evidence versions are independently snapshotted, including setup, timer and ordered-angle failures, interrupted trials, the failed V25 gate and the complete V26 development. The latter cannot inherit independent status. Raw records, protocols, source hashes and generated tables are preserved in `attempts/49-rational-grammar-drift/work`; model weights/caches are excluded. No result is inferred from a table cap or changed control implementation.

The geometry checks include 1,860 cases, 33,747 exact norm directions, 117,376 projections by independently scanning every segment, 1,860 ordered image comparisons, 1,167 products and 389 at $\varepsilon = 2^{-1024}$. Original-ID forest cases use independent JSON/stack recognizers and direct assignment enumeration. Known-root verification covers all 300 old word/engine cases; classical two-sided controls cover 105 cases including 33 undefined-zero-mass refusals. The 6 proof-family checks use an exact binomial counter. These are correctness checks, not publication evidence. The maintained offline suite passes all 170 tests, including the original-ID known-root regression. Selected Lean statements elsewhere do not verify this complete geometric theorem, grammar compiler or Python implementation.

Run the ordered oracle with a fresh `-output` directory:

```
python -m
attempts.49-rational-grammar-drift.work.verify_ordered
```

The external campaign module is:

```
attempts.49-rational-grammar-drift.work.audit_external
```

V26 explicitly uses `-known-root` and the same stored V25 capture. Table 1 is generated by `build_report_v27`, verifying both immutable record inventories and decision hashes before deriving all45 pairs. The old public packet is available offline; the new15 raw heads require the optional local capture inputs, not just their hashes. Exact-softmax replay runs no forward and does not independently reproduce neural logits from weights.

Independent human checking of the theorem and novelty/significance remains absent. The improved quantitative upper bound is mathematically separate from physical timing: the independent practical precision gate failed, and one new developmental diagnostic does not establish a broad useful gain. The proposed use is deterministic numerical-fidelity auditing with explicit costs and limitations. Native EPIC, learned semantic correctness, faster complete generation and publication acceptance remain outside the proved claim. The mathematical argument should be assessed on its actual quantitative advance, rather than on code size, number of trials or successful examples.

References

[1] Y. Fu. On Deterministically Computing Total Variation Distance via Zonotope Compression. arXiv:2609.24235v1, 2026. https://arxiv.org/abs/2609.24235v1.

[2] W. Feng, L. Liu and T. Liu. On Deterministically Approximating Total Variation Distance. SODA, 1766–1791, 2024; arXiv:2309.14696v1. https://arxiv.org/abs/2309.14696v1.

[3] S. Har-Peled and M. Jones. A Proof of Dudley's Convex Approximation. Manuscript, 2017, revision May 2024. https://sarielhp.org/p/17/dudley_approx/dudley_approx.pdf.

[4] J. Matoušek. Improved Upper Bounds for Approximation by Zonotopes. Acta Mathematica 177:55–73, 1996. https://doi.org/10.1007/BF02392598.

[5] A. Bhattacharyya, S. Gayen, K. S. Meel, D. Myrisiotis, A. Pavan and N. V. Vinodchandran. Total Variation Distance Meets Probabilistic Inference. ICML, PMLR 235:3776–3794, 2024. https://proceedings.mlr.press/v235/bhattacharyya24a.html.

[6] A. Bhattacharyya, G. Cormode, Y. Fu and K. S. Meel. Total Variation Distance Estimation through Domain Reduction. arXiv:2609.18707v2, 2026. https://arxiv.org/abs/2609.18707v2.

[7] K. Anand, A. Benford and H. Guo. Linear Time Approximation of the Total Variation Distance between Product Distributions. arXiv:2607.27088, 2026. https://arxiv.org/abs/2607.27088.

[8] A. Bhattacharyya, W. Feng and P. Srivastava. Approximating the Total Variation Distance between Gaussians. COLT, PMLR 258, 2025. https://proceedings.mlr.press/v258/bhattacharyya25a.html.

[9] S. S. Sahoo, M. Arriola, Y. Schiff, A. Gokaslan, E. Marroquin, J. T. Chiu, A. Rush and V. Kuleshov. Simple and Effective Masked Diffusion Language Models. NeurIPS 37, 2024. https://arxiv.org/abs/2406.07524.

[10] J. Goodman. Semiring Parsing. Computational Linguistics 25(4):573–605, 1999. https://aclanthology.org/J99-4004/.

[11] S. Rivaud and F. Pachet. Sampling Markov Models under Constraints: Complexity Results for Binary Equalities and Grammar Membership. arXiv:1711.10436, 2017. https://arxiv.org/abs/1711.10436.

[12] H. Jin and Y.-S. Han. EPIC: Efficient and Parallel Inference under CFG Constraints for Diffusion Language Models. arXiv:2606.00722v2, 2026. https://arxiv.org/abs/2606.00722v2.